Run report: item2P-run-20261018T2056Z

Descriptive record of one recorded run of examples/pkm_coverage.rs (Part P) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: item2P, 20261018T2056Z
protocol commits: 55ebe41b6 (Addendum A), Amendment B (Poisson residual, frozen before this run); tool commit: fe0469c07
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz, 15 GB
command: rustc --edition 2021 -O examples/pkm_coverage.rs -o tool && PKM_THREADS=2 ./tool research/weil_descent_angles_20261018/results/item2P-run-20261018T2056Z
tool sha256: 85bda43eecd9e476bf74c4cc9d9ad86d444d42be7a9e9169b6c0b2dbcf08a254
threads: 2 (sharing the host with the item-4 census and the other coverage run; outputs are exact counts, no timing is evidence)
smoke runs before this run (not evidence): DC_SCALE=0.1 / PKM_SCALE=0.15, single thread.
```

tool_log.txt

```
jobs: 144
P kummer m=2 t_off=-1: primes 24 pi family 0.2158 null 0.2088 positive 0.0022 | resid family -0.00024 null +0.00000 positive -0.2138 | paired resid diff -0.00024 (z -1.10)
P kummer m=2 t_off=+0: primes 24 pi family 0.5793 null 0.5920 positive 0.0044 | resid family +0.00034 null +0.00075 positive -0.5746 | paired resid diff -0.00041 (z -1.40)
P kummer m=2 t_off=+1: primes 24 pi family 0.9501 null 0.9501 positive 0.0087 | resid family +0.00013 null +0.00037 positive -0.9412 | paired resid diff -0.00024 (z -1.84)
P kummer m=3 t_off=-1: primes 24 pi family 0.1168 null 0.1208 positive 0.0013 | resid family -0.00731 null -0.00685 positive -0.1228 | paired resid diff -0.00046 (z -0.43)
P kummer m=3 t_off=+0: primes 24 pi family 0.5884 null 0.5873 positive 0.0029 | resid family -0.00919 null -0.00962 positive -0.5947 | paired resid diff +0.00043 (z 0.26)
P kummer m=3 t_off=+1: primes 24 pi family 0.9910 null 0.9892 positive 0.0058 | resid family -0.00030 null -0.00073 positive -0.9855 | paired resid diff +0.00043 (z 1.36)
P dickson_split m=2 t_off=-1: primes 24 pi family 0.2356 null 0.2291 positive 0.0029 | resid family -0.00040 null -0.00049 positive -0.2331 | paired resid diff +0.00008 (z 0.34)
P dickson_split m=2 t_off=+0: primes 24 pi family 0.6455 null 0.6464 positive 0.0058 | resid family +0.00065 null +0.00051 positive -0.6391 | paired resid diff +0.00015 (z 0.39)
P dickson_split m=2 t_off=+1: primes 23 pi family 0.9740 null 0.9740 positive 0.0116 | resid family +0.00027 null +0.00032 positive -0.9621 | paired resid diff -0.00004 (z -0.34)
P dickson_split m=3 t_off=-1: primes 24 pi family 0.1194 null 0.1315 positive 0.0014 | resid family -0.00782 null -0.00729 positive -0.1259 | paired resid diff -0.00053 (z -0.85)
P dickson_split m=3 t_off=+0: primes 24 pi family 0.5945 null 0.6123 positive 0.0030 | resid family -0.00929 null -0.01165 positive -0.6009 | paired resid diff +0.00236 (z 1.71)
P dickson_split m=3 t_off=+1: primes 24 pi family 0.9812 null 0.9823 positive 0.0058 | resid family -0.00109 null -0.00101 positive -0.9765 | paired resid diff -0.00008 (z -0.22)
P dickson_torus m=2 t_off=-1: primes 24 pi family 0.2231 null 0.2258 positive 0.0021 | resid family -0.00034 null -0.00021 positive -0.2214 | paired resid diff -0.00013 (z -0.53)
P dickson_torus m=2 t_off=+0: primes 23 pi family 0.6174 null 0.6194 positive 0.0041 | resid family +0.00037 null +0.00042 positive -0.6130 | paired resid diff -0.00005 (z -0.23)
P dickson_torus m=2 t_off=+1: primes 24 pi family 0.9636 null 0.9638 positive 0.0084 | resid family +0.00026 null +0.00031 positive -0.9550 | paired resid diff -0.00005 (z -0.45)
P dickson_torus m=3 t_off=-1: primes 24 pi family 0.1449 null 0.1185 positive 0.0017 | resid family -0.00698 null -0.00684 positive -0.1502 | paired resid diff -0.00014 (z -0.16)
P dickson_torus m=3 t_off=+0: primes 24 pi family 0.6038 null 0.6119 positive 0.0035 | resid family -0.00931 null -0.00831 positive -0.6096 | paired resid diff -0.00100 (z -0.48)
P dickson_torus m=3 t_off=+1: primes 24 pi family 0.9921 null 0.9891 positive 0.0071 | resid family -0.00044 null -0.00061 positive -0.9855 | paired resid diff +0.00018 (z 0.71)
total 257.2s
```

checks.txt

```
P kummer m=2 t_off=-1: primes 24 pi family 0.2158 null 0.2088 positive 0.0022 | resid family -0.00024 null +0.00000 positive -0.2138 | paired resid diff -0.00024 (z -1.10)
P kummer m=2 t_off=+0: primes 24 pi family 0.5793 null 0.5920 positive 0.0044 | resid family +0.00034 null +0.00075 positive -0.5746 | paired resid diff -0.00041 (z -1.40)
P kummer m=2 t_off=+1: primes 24 pi family 0.9501 null 0.9501 positive 0.0087 | resid family +0.00013 null +0.00037 positive -0.9412 | paired resid diff -0.00024 (z -1.84)
P kummer m=3 t_off=-1: primes 24 pi family 0.1168 null 0.1208 positive 0.0013 | resid family -0.00731 null -0.00685 positive -0.1228 | paired resid diff -0.00046 (z -0.43)
P kummer m=3 t_off=+0: primes 24 pi family 0.5884 null 0.5873 positive 0.0029 | resid family -0.00919 null -0.00962 positive -0.5947 | paired resid diff +0.00043 (z 0.26)
P kummer m=3 t_off=+1: primes 24 pi family 0.9910 null 0.9892 positive 0.0058 | resid family -0.00030 null -0.00073 positive -0.9855 | paired resid diff +0.00043 (z 1.36)
P dickson_split m=2 t_off=-1: primes 24 pi family 0.2356 null 0.2291 positive 0.0029 | resid family -0.00040 null -0.00049 positive -0.2331 | paired resid diff +0.00008 (z 0.34)
P dickson_split m=2 t_off=+0: primes 24 pi family 0.6455 null 0.6464 positive 0.0058 | resid family +0.00065 null +0.00051 positive -0.6391 | paired resid diff +0.00015 (z 0.39)
P dickson_split m=2 t_off=+1: primes 23 pi family 0.9740 null 0.9740 positive 0.0116 | resid family +0.00027 null +0.00032 positive -0.9621 | paired resid diff -0.00004 (z -0.34)
P dickson_split m=3 t_off=-1: primes 24 pi family 0.1194 null 0.1315 positive 0.0014 | resid family -0.00782 null -0.00729 positive -0.1259 | paired resid diff -0.00053 (z -0.85)
P dickson_split m=3 t_off=+0: primes 24 pi family 0.5945 null 0.6123 positive 0.0030 | resid family -0.00929 null -0.01165 positive -0.6009 | paired resid diff +0.00236 (z 1.71)
P dickson_split m=3 t_off=+1: primes 24 pi family 0.9812 null 0.9823 positive 0.0058 | resid family -0.00109 null -0.00101 positive -0.9765 | paired resid diff -0.00008 (z -0.22)
P dickson_torus m=2 t_off=-1: primes 24 pi family 0.2231 null 0.2258 positive 0.0021 | resid family -0.00034 null -0.00021 positive -0.2214 | paired resid diff -0.00013 (z -0.53)
P dickson_torus m=2 t_off=+0: primes 23 pi family 0.6174 null 0.6194 positive 0.0041 | resid family +0.00037 null +0.00042 positive -0.6130 | paired resid diff -0.00005 (z -0.23)
P dickson_torus m=2 t_off=+1: primes 24 pi family 0.9636 null 0.9638 positive 0.0084 | resid family +0.00026 null +0.00031 positive -0.9550 | paired resid diff -0.00005 (z -0.45)
P dickson_torus m=3 t_off=-1: primes 24 pi family 0.1449 null 0.1185 positive 0.0017 | resid family -0.00698 null -0.00684 positive -0.1502 | paired resid diff -0.00014 (z -0.16)
P dickson_torus m=3 t_off=+0: primes 24 pi family 0.6038 null 0.6119 positive 0.0035 | resid family -0.00931 null -0.00831 positive -0.6096 | paired resid diff -0.00100 (z -0.48)
P dickson_torus m=3 t_off=+1: primes 24 pi family 0.9921 null 0.9891 positive 0.0071 | resid family -0.00044 null -0.00061 positive -0.9855 | paired resid diff +0.00018 (z 0.71)
```

pkm_coverage.csv (first 12 of 1290 rows)

family	m	p	r	log2r	t	t_offset	arm	V	F	heuristic	lambda	poisson	distinct	multisets	pi

kummer	2	65537	65579	16.001	8	−1	kummer	256	264	0.499672	0.533402	0.413394	27109	34980	0.413379
kummer	2	65537	65579	16.001	8	−1	null	256	278	0.499672	0.591363	0.446428	29231	38781	0.445737
kummer	2	65537	65579	16.001	8	−1	positive	256	264	0.499672	0.533402	0.413394	529	34980	0.008067
kummer	2	65537	65579	16.001	9	0	kummer	512	488	1.998689	1.819424	0.837881	54915	119316	0.837387
kummer	2	65537	65579	16.001	9	0	null	512	518	1.998689	2.049757	0.871234	57397	134421	0.875234
kummer	2	65537	65579	16.001	9	0	positive	512	488	1.998689	1.819424	0.837881	977	119316	0.014898
kummer	2	65537	65579	16.001	10	1	kummer	1024	1040	7.994754	8.254472	0.999740	65555	541320	0.999634
kummer	2	65537	65579	16.001	10	1	null	1024	1012	7.994754	7.816191	0.999597	65565	512578	0.999787
kummer	2	65537	65579	16.001	10	1	positive	1024	1040	7.994754	8.254472	0.999740	2081	541320	0.031733
kummer	2	70657	71089	16.117	8	−1	kummer	256	274	0.460943	0.529969	0.411377	29141	37675	0.409923
kummer	2	70657	71089	16.117	8	−1	null	256	256	0.460943	0.462744	0.370446	26359	32896	0.370789
kummer	2	70657	71089	16.117	8	−1	positive	256	274	0.460943	0.529969	0.411377	549	37675	0.007723

pkm_summary.csv (first 12 of 18 rows)

family	m	t_offset	primes	mean_pi_family	mean_pi_null	mean_pi_positive	resid_family	resid_null	resid_positive	resid_diff	sd_resid_diff	z_resid_diff
kummer	2	−1	24	0.215779	0.208832	0.002236	−0.000240	0.000002	−0.213782	−0.000242	0.001083	−1.10
kummer	2	0	24	0.579288	0.592036	0.004368	0.000344	0.000753	−0.574576	−0.000409	0.001430	−1.40
kummer	2	1	24	0.950070	0.950146	0.008726	0.000132	0.000373	−0.941212	−0.000240	0.000640	−1.84
kummer	3	−1	24	0.116842	0.120809	0.001339	−0.007311	−0.006851	−0.122813	−0.000460	0.005279	−0.43
kummer	3	0	24	0.588405	0.587274	0.002926	−0.009194	−0.009619	−0.594672	0.000425	0.008148	0.26
kummer	3	1	24	0.990996	0.989248	0.005817	−0.000298	−0.000727	−0.985477	0.000429	0.001547	1.36
dickson_split	2	−1	24	0.235561	0.229063	0.002877	−0.000403	−0.000487	−0.233087	0.000084	0.001232	0.34
dickson_split	2	0	24	0.645538	0.646389	0.005778	0.000655	0.000505	−0.639105	0.000149	0.001890	0.39
dickson_split	2	1	23	0.974048	0.973990	0.011647	0.000273	0.000316	−0.962128	−0.000043	0.000609	−0.34
dickson_split	3	−1	24	0.119446	0.131493	0.001405	−0.007817	−0.007290	−0.125858	−0.000527	0.003037	−0.85
dickson_split	3	0	24	0.594514	0.612283	0.002950	−0.009292	−0.011650	−0.600856	0.002358	0.006771	1.71
dickson_split	3	1	24	0.981202	0.982349	0.005770	−0.001093	−0.001012	−0.976526	−0.000082	0.001782	−0.22

Graph

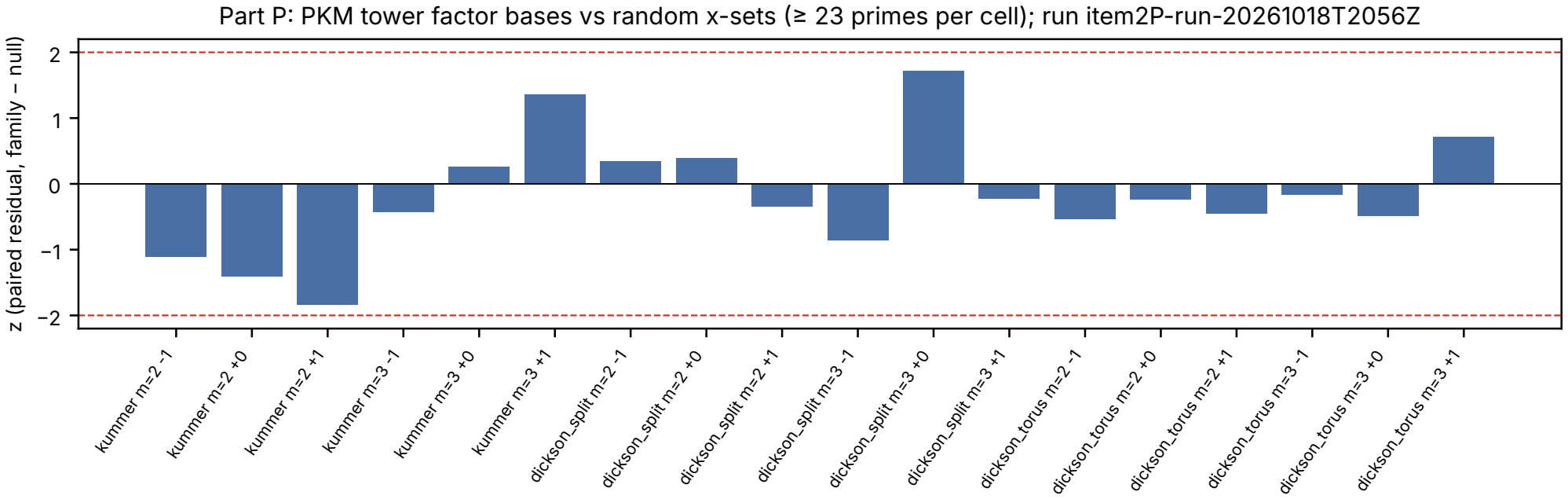


Figure 1: Part P: z of the paired Poisson residual, tower factor base vs random x-set, per cell.